Satoshi TAJIMA, PhD

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EDUCATION	**PhD** in Environmental Sciences The University of Tokyo Tokyo, Japan *Thesis title: "Seawater-groundwater interactions in highly permeable coastal aquifers"* *Advisor: Prof Tomochika Tokunaga*	**March 2025**
	MSc in Environmental Sciences The University of Tokyo Tokyo, Japan	**March 2022**
	Bachelor of Agriculture The University of Tokyo Tokyo, Japan	**March 2020**
PROFESSIONAL EXPERIENCE	**Lecturer *(Chargé d'enseignement)*** Centre for Hydrogeology and Geothermics, University of Neuchâtel Neuchâtel, Switzerland	**August 2025 –**
	Postdoctoral Researcher Centre for Hydrogeology and Geothermics, University of Neuchâtel Neuchâtel, Switzerland	**April 2025 –**
	Postdoctoral Researcher Physics Institute, University of Bern Bern, Switzerland	**April 2025 –**
	Visiting Researcher Graduate School of Frontier Sciences, The University of Tokyo Kashiwa, Japan	**December 2025 –**
	Research Fellow Japan Society for the Promotion of Science Tokyo, Japan	**April 2023 –** **March 2025**
	Visiting Researcher Institute of Environmental Assessment and Water Research, Spanish National Research Council (IDAEA-CSIC) Barcelona, Spain	**September 2022 –** **September 2023**
	Research Assistant WINGS-PES, The University of Tokyo Kashiwa, Japan	**October 2020 –** **March 2023**
PEER-REVIEWED JOURNAL ARTICLES	**Tajima, S.**, Therrien, R., Brunner, P. (2026). Climate change alters post-surge recovery of coastal aquifers. *Water Resources Research,* **62**(3), e2025WR042142.	

Tajima, S., Dentz, M. (2025). Tides and heterogeneity drive trapping and chaotic mixing in coastal aquifers. *Water Resources Research,* **61**(11), e2025WR040620.

Tajima, S., Dentz, M. (2025). Transient forcing in heterogeneous aquifers drives solute containment and chaotic mixing. *Advances in Water Resources*, **203**, 105021.

Tajima, S., Dentz, M., Liu, J., Tokunaga, T. (2024). Time-dependent dispersion coefficients for the evolution of displacement fronts in heterogeneous porous media. *Advances in Water Resources*, **189**, 104714.

Tajima, S., Yamaguchi, K., Shiroyama, H. (2024). Consensus building with fishermen on offshore wind farms in Japan: Current status and policy recommendations. *Marine Policy*, **160**, 105975.

Tajima, S., Liu, J., Tokunaga, T. (2024). Mechanisms of non-fresh groundwater presence at water tables in highly permeable coastal aquifers. *Groundwater*, **62**(4), 527–536.

Tajima, S., Brunner, P., Liu, J., Delottier, H., Tokunaga, T. (2023). Groundwater flooding on atolls caused by storm surges: Effects of the dual-aquifer configuration. *Water Resources Research,* **59**(10), e2023WR034762.

Tajima, S., Liu, J., Tokunaga, T. (2022). Estimating freshwater lens volume in highly permeable aquifers. *Hydrological Research Letters*, **16**(1), 12–17.

Tajima, S., Yoshida, S., Fukui, T., Nihei, N., Kobayashi, N. I. (2022). Cesium-137 stored on and discharged from banks of an agricultural canal in Iitate, Fukushima. *Journal of Environmental Radioactivity*, **241**, 106775.

ARTICLES IN JAPANESE (Peer-reviewed)

Yamaguchi, K., **Tajima, S.**, Katsuno, T., Shiroyama, H. (2024). Challenges and proposals for regional community transitions with offshore wind projects: A case study in Akita Prefecture, Japan. *Environmental Information Science*, **25**(2), 55–64.
(洋上風力発電をめぐる地域社会の移行における課題と提言 —秋田県を事例に—, *環境情報科学*, **25**(2), 55–64.)

Yamaguchi, K., **Tajima, S.**, Watanabe, A., Shiroyama, H. (2024). Consensus building with fishermen in Japanese offshore wind projects: a case study in the Oga-Katagami-Akita district and policy recommendations. *Japan Society of Ocean Policy*, **13**, 65–81.
(我が国の洋上風力事業における漁業者との合意形成: 秋田県男鹿市、潟上市及び秋田市沖における事例と政策提言, *日本海洋政策学会誌*, **13**, 65–81.)

Tajima, S., Otori, H., Yamaguchi, K. (2021). Contribution of offshore windfarm projects to regional economies: base ports in Akita Prefecture. *Journal of Japan Society of Ocean Policy*, **11**, 27–41.
(洋上風力産業の地域経済への貢献: 秋田県における拠点港の役割. *日本海洋政策学会誌*, **11**, 27–41.)

BOOKS

Bosch. T., Colijn, F. Ebinghaus, R., Körtzinger, A., Latif, M., …, Weinberger, F. (2022). *World Ocean Review 1* (Ocean Alliance Collaborative Research Organization, The University of Tokyo, Trans.), maribus. (Original work published 2010) (Contributed as co-translator)

AWARDS & HONOURS

Dean's Award for Outstanding Achievement (Doctoral Course) **March 2025**
Graduate School of Frontier Sciences,
The University of Tokyo

	Outstanding Student Presentation Award Japan Geoscience Union	**July 2023**
	Award of Accomplishment **Grants-in-Aid of Groundwater Research for Young Investigators** Japanese Association of Groundwater Hydrology	**August 2022**
	Dean's Award for Outstanding Achievement (Master Course) Graduate School of Frontier Sciences, The University of Tokyo	**March 2022**
	Outstanding Presentation Award for Young Researcher Japanese Association of Groundwater Hydrology	**December 2021**
FELLOWSHIPS	**JSPS Research Fellowship for Young Scientists (DC2)** Japan Society for the Promotion of Science	**April 2023 – March 2025**
	WINGS-PES Fellowship World-leading Innovative Graduate Study Program in Proactive Environmental Studies (WINGS-PES), The University of Tokyo	**October 2020 – March 2023**
COMPETITIVE RESEARCH GRANTS	**Grant-in-Aid for JSPS Research Fellows** ▪ Applicant: **Tajima, S.** ▪ Title: Establishing a new subsurface mass transport model to mitigate upstream dispersion Funding: JPY 2,800,000	**April 2023 – March 2025**
	GSFS Challenging New Area Doctoral Research Grant, The University of Tokyo ▪ Applicants: **Tajima, S.**, Ito, M., Nakagawa, Y. ▪ Title: Elucidating groundwater environmental conditions under which the uncultivated bacterial group CPR dominates ▪ Funding: JPY 600,000	**June 2022 – May 2023**
	Tokyo Geographical Society Research Grant ▪ Applicant: **Tajima, S.** ▪ Title: Illustrating Salinity distribution in the unsaturated zone on Tarama Island, Japan ▪ Funding: JPY 500,000	**June 2022 – May 2023**
	JSPS Bilateral Joint Research Project ▪ Applicants: Tokunaga, T., Aichi, M., Liu, J., Ito, Y., Ouyang, M., **Tajima, S.**, Brunner, P., Schilling, O. ▪ Title: The impact of coastal flooding induced saline water infiltration into subsurface in urban areas:	**April 2021 – March 2023**

Integrating spatially high-resolution surface processes and density-driven groundwater flow models

- Funding: JPY 3,880,000

Grant-in-Aid of Groundwater Research for Young Investigators, Japanese Association of Groundwater Hydrology — **July 2021 – June 2022**

- Applicant: **Tajima, S.**
- Title: Assessment of climate change impacts on freshwater lenses of raised coral reef islands
- Funding: JPY 250,000

COURSE TEACHING

Course Instructor – "Hydrogeological Processes" — **September 2026 – January 2027**
Faculty of Science, University of Neuchâtel
Autumn Semester 2026/27 (3 ECTS credits)

Teaching Assistant – "Seminar on Marine Affairs" — **October 2021 – January 2025**
Graduate School of Public Policy, The University of Tokyo
Autumn Semester 2021/22
Spring/Autumn Semesters 2022/23
Spring/Autumn Semesters 2023/24
Spring/Autumn Semesters 2024/25 (4 credits)

Teaching Assistant – "Kashiwa Science Camp" — **January 2022 – March 2022**
Graduate School of Frontier Sciences, The University of Tokyo
Intensive Course 2021/22 (1 credit)

Teaching Assistant – "Marine Science and Technology Policy" — **September 2021 – January 2022 / September 2022 – January 2023**
Graduate School of Public Policy, The University of Tokyo
Autumn Semester 2021/22
Autumn Semester 2022/23 (2 credits)

GUEST LECTURES

Department of Water Resources, Republic of Vanuatu — **May 2026 / August 2026**
"Introduction to Geophysical Survey"
"Introduction to Hydrology and Hydrogeology: Seawater Intrusion"

École Japonaise Complémentaire de Genève — **November 2025**
"Water Cycle and Water Resources"

Akita Prefectural University — **November 2024**
"The Relationship between Offshore Wind and Local Communities"

ACADEMIC SERVICES

Local Scientific Committee Member — **September 2026 –**
29th Salt Water Intrusion Meeting (SWIM)

Session Convenor — **May 2026**
JpGU-AGU Joint Meeting 2026
"Surface Water-Groundwater Interactions: Physical, Chemical, and Biological Processes"

Session Convenor — **May 2025**

JpGU Meeting 2025
“Hydrological processes of surface-groundwater interactions”

JOURNAL PEER-REVIEWS

- *Advances in Water Resources*
- *Environmental Engineering and Management Journal*
- *Hydrogeology Journal*
- *Hydrological Processes*
- *Journal of Hydrology: Regional Studies*
- *Marine Policy*
- *Scientific Data*
- *Scientific Reports*
- *Soil Science Society of America Journal*
- *Transport in Porous Media*
- *Water Resources Research*

PROFESSIONAL MEMBERSHIPS

- European Geosciences Union (EGU)
- International Association of Hydrogeologists (IAH)
  - IAH Japan Chapter
  - Early Career Hydrogeologists' Network (ECHN)
  - Coastal Groundwater Network (CGN)
- Japan Geoscience Union (JpGU)
- Japanese Association of Groundwater Hydrology

LANGUAGES

- Japanese (native)
- English (full professional proficiency)
- French / German (limited professional proficiency)
- Spanish / Russian (elementary)